

ADVANCED

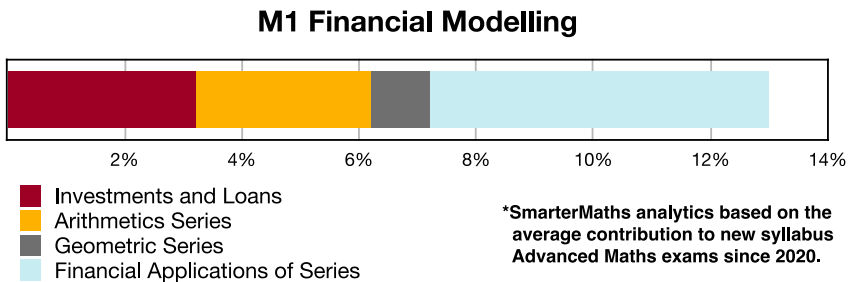
Stage 6

Financial Maths (Adv), M1 Financial Modelling (Adv)

Financial Applications of Series (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 177 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- M1 Financial Modelling* has contributed a substantial 13.0% to new syllabus Advanced exams since it was introduced in 2020.
- The topic is split into four sub-topics for the purposes of analysis: 1-*Investments and Loans* (3.2%), 2-*Arithmetic Series* (3.0%), 3-*Geometric Series* (1.0%) and 4-*Financial Applications of Series* (5.8%).
- This analysis looks at *Financial Applications of Series*.

HSC ANALYSIS - What to expect and common pitfalls

- Financial Applications of Series* is examined like clockwork in the new syllabus *Advanced* exam, accounting for a significant 5.8% of the exam.
- Students need to be confident in dealing with 3 main question types: 1-Loans, 2-Annuity (growth phase), and 3-Annuity (withdrawal phase).
- 1-Loans were last examined in 2022 after a six year absence.
- 2-Annuity (growth phase) was last examined in 2024. Examiners have included a difficult twist involving increasing annuity payments that is important revision (review 2014 Adv 16b).
- 3-Annuity (withdrawal phase) represents the upper difficulty of this topic area, examined 5 times in 6 years between 2018-23. 2020 Q26 involved an annuity balance expressed as a "recurrence equation" for the 1st time, and is important revision.
- Pitfalls*: Marker's Comments - good setting out and clear handwriting is at a premium with messy working a major contributor to errors. Best practice is illustrated in worked solutions - take note!

Questions

1. Financial Maths, 2ADV M1 2016 HSC 14b

A gardener develops an eco-friendly spray that will kill harmful insects on fruit trees without contaminating the fruit. A trial is to be conducted with 100 000 insects. The gardener expects the spray to kill 35% of the insects each day and that exactly 5000 new insects will be produced each day.

The number of insects expected at the end of the n th day of the trial is A_n .

- Show that $A_2 = 0.65(0.65 \times 100\,000 + 5000) + 5000$. (2 marks)
- Show that $A_n = 0.65^n \times 100\,000 + 5000 \frac{(1 - 0.65^n)}{0.35}$. (1 mark)
- Find the expected insect population at the end of the fourteenth day, correct to the nearest 100. (1 mark)

2. Financial Maths, 2ADV M1 2020 HSC 26

Tina inherits \$60 000 and invests it in an account earning interest at a rate of 0.5% per month. Each month, immediately after the interest has been paid, Tina withdraws \$800.

The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.005) - 800,$$

where $n = 1, 2, 3, \dots$ and $A_0 = 60\,000$

- Use the recurrence relation to find the amount of money in the account immediately after the third withdrawal. (2 marks)
- Calculate the amount of interest earned in the first three months. (2 marks)
- Calculate the amount of money in the account immediately after the 94th withdrawal. (3 marks)

3. Financial Maths, 2ADV M1 2009 HSC 8b

One year ago Daniel borrowed \$350 000 to buy a house. The interest rate was 9% per annum, compounded monthly. He agreed to repay the loan in 25 years with equal monthly repayments of \$2937.

- Calculate how much Daniel owed after his first monthly repayment. (1 mark)

Daniel has just made his 12th monthly repayment. He now owes \$346 095. The interest rate now decreases to 6% per annum, compounded monthly.

The amount $\$A_n$, owing on the loan after the n th monthly repayment is now calculated using the formula

$$A_n = 346,095 \times 1.005^n - 1.005^{n-1}M - \dots - 1.005M - M$$

- where $\$M$ is the monthly repayment, and $n = 1, 2, \dots, 288$. (DO NOT prove this formula.)
- Calculate the monthly repayment if the loan is to be repaid over the remaining 24 years (288 months). (3 marks)
 - Daniel chooses to keep his monthly repayments at \$2937. Use the formula in part (ii) to calculate how long it will take him to repay the \$346 095. (3 marks)
 - How much will Daniel save over the term of the loan by keeping his monthly repayments at \$2937, rather than reducing his repayments to the amount calculated in part (ii)? (1 mark)

4. Financial Maths, 2ADV M1 2015 HSC 14c

Sam borrows \$100 000 to be repaid at a reducible interest rate of 0.6% per month. Let $\$A_n$ be the amount owing at the end of n months and $\$M$ be the monthly repayment.

i. Show that $A_2 = 100\,000(1.006)^2 - M(1 + 1.006)$. (1 mark)

ii. Show that $A_n = 100\,000(1.006)^n - M\left(\frac{(1.006)^n - 1}{0.006}\right)$. (2 marks)

iii. Sam makes monthly repayments of \$780. Show that after making 120 monthly repayments the amount owing is \$68 500 to the nearest \$100. (1 mark)

Immediately after making the 120th repayment, Sam makes a one-off payment, reducing the amount owing to \$48 500. The interest rate and monthly repayment remain unchanged.

iv. After how many more months will the amount owing be completely repaid? (3 marks)

5. Financial Maths, 2ADV M1 2021 HSC 29

a. On the day that Megan was born, her grandfather deposited \$5000 into an account earning 3% per annum compounded annually. On each birthday after this, her grandfather deposited \$1000 into the same account, making his final deposit on Megan's 17th birthday. That is, a total of 18 deposits were made.

Let A_n be the amount in the account on Megan's n th birthday, after the deposit is made.

Show that $A_3 = \$8554.54$. (2 marks)

b. On her 17th birthday, just after the final deposit is made, Megan has \$30 025.83 in her account. You are NOT required to show this.

Megan then decides to leave all the money in the same account continuing to earn interest at 3% per annum compounded annually. On her 18th birthday, and on each birthday after this, Megan withdraws \$2000 from the account.

How many withdrawals of \$2000 will Megan be able to make? (3 marks)

6. Financial Maths, 2ADV M1 2023 HSC 25

On the first day of November, Jia deposits \$10 000 into a new account which earns 0.4% interest per month, compounded monthly. At the end of each month, after the interest is added to the account, Jia intends to withdraw $\$M$ from the account.

Let A_n be the amount (in dollars) in Jia's account at the end of n months.

a. Show that $A_2 = 10\,000(1.004)^2 - M(1.004) - M$. (1 mark)

b. Show that $A_n = (10\,000 - 250M)(1.004)^n + 250M$. (3 marks)

c. Jia wants to be able to make at least 100 withdrawals.

What is the largest value of M that will enable Jia to do this? (2 marks)

7. Financial Maths, 2ADV M1 2024 HSC 24

Jack intends to deposit \$80 into a savings account on the first day of each month for 24 months. The interest rate during this time is 6% per annum, compounded monthly.

a. Calculate how much money Jack will have in his account at the end of the 24 months. (3 marks)

b. A table of future value interest factors could have been used to calculate how much money Jack would have in his account at the end of the 24 months.

Part of the table is shown.

Table of future value interest factors for an annuity of \$1

Number of periods	Interest rate per period		
		0.5%	
24		A	

What is the value of A in the table? Give your answer correct to 3 decimal places. (1 mark)

8. Financial Maths, 2ADV M1 2010 HSC 9a

i. When Chris started a new job, \$500 was deposited into his superannuation fund at the beginning of each month. The money was invested at 0.5% per month, compounded monthly.

Let $\$P$ be the value of the investment after 240 months, when Chris retires.

Show that $P = 232\,175.55$ (2 marks)

ii. After retirement, Chris withdraws \$2000 from the account at the end of each month, without making any further deposits. The account continues to earn interest at 0.5% per month.

Let $\$A_n$ be the amount left in the account n months after Chris's retirement.

(1) Show that $A_n = (P - 400\,000) \times 1.005^n + 400\,000$. (3 marks)

(2) For how many months after retirement will there be money left in the account? (2 marks)

9. Financial Maths, 2ADV M1 2011 HSC 8c

- When Jules started working she began paying \$100 at the beginning of each month into a superannuation fund.
- The contributions are compounded monthly at an interest rate of 6% per annum.
- She intends to retire after having worked for 35 years.
- i. Let $\$P$ be the final value of Jules's superannuation when she retires after 35 years (420 months). Show that $\$P = \$143\,183$ to the nearest dollar. *(2 marks)*
- ii. Fifteen years after she started working Jules read a magazine article about retirement, and realised that she would need \$800 000 in her fund when she retires. At the time of reading the magazine article she had \$29 227 in her fund. For the remaining 20 years she intends to work, she decides to pay $\$M$ into her fund at the beginning of each month. The contributions continue to attract the same interest rate of 6% per annum, compounded monthly.

At the end of n months after starting the new contributions, the amount in the fund is $\$A_n$.

- (1) Show that $A_2 = 29\,227 \times 1.005^2 + M(1.005 + 1.005^2)$. *(1 mark)*
- (2) Find the value of M so that Jules will have \$800 000 in her fund after the remaining 20 years (240 months). *(3 marks)*

10. Financial Maths, 2ADV M1 2012 HSC 15c

- Ari takes out a loan of \$360 000. The loan is to be repaid in equal monthly repayments, $\$M$, at the end of each month, over 25 years (300 months). Reducible interest is charged at 6% per annum, calculated monthly.
- Let $\$A_n$ be the amount owing after the n th repayment.
- i. Write down an expression for the amount owing after two months, $\$A_2$. *(1 mark)*
- ii. Show that the monthly repayment is approximately \$2319.50. *(2 marks)*
- iii. After how many months will the amount owing, $\$A_n$, become less than \$180 000. *(3 marks)*

11. Financial Maths, 2ADV M1 2004 HSC 7c

- Betty decides to set up a trust fund for her grandson, Luis. She invests \$80 at the beginning of each month. The money is invested at 6% per annum, compounded monthly.
- The trust fund matures at the end of the month of her final investment, 25 years after her first investment. This means that Betty makes 300 monthly investments.
- i. After 25 years, what will be the value of the first \$80 invested? *(2 marks)*
- ii. By writing a geometric series for the value of all Betty's investments, calculate the final value of Luis' trust fund. *(3 marks)*

12. Financial Maths, 2ADV M1 2005 HSC 8c

- Weelabarrabak Shire Council borrowed \$3 000 000 at the beginning of 2005. The annual interest rate is 12%. Each year, interest is calculated on the balance at the beginning of the year and added to the balance owing. The debt is to be repaid by equal annual repayments of \$480 000, with the first repayment being made at the end of 2005.
- Let A_n be the balance owing after the n -th repayment.
- i. Show that $A_2 = (3 \times 10^6)(1.12)^2 - (4.8 \times 10^5)(1 + 1.12)$. *(1 mark)*
- ii. Show that $A_n = 10^6[4 - (1.12)^n]$. *(2 marks)*
- iii. In which year will Weelabarrabak Shire Council make the final repayment? *(2 marks)*

13. Financial Maths, 2ADV M1 2006 HSC 8b

- Joe borrows \$200 000 which is to be repaid in equal monthly instalments. The interest rate is 7.2% per annum reducible, calculated monthly.
- It can be shown that the amount, $\$A_n$, owing after the n th repayment is given by the formula:
- $$A_n = 200\,000r^n - M(1 + r + r^2 + \dots + r^{n-1}),$$
- where $r = 1.006$ and $\$M$ is the monthly repayment. (Do NOT show this.)
- i. The minimum monthly repayment is the amount required to repay the loan in 300 instalments. Find the minimum monthly repayment. *(3 marks)*
- ii. Joe decides to make repayments of \$2800 each month from the start of the loan. How many months will it take for Joe to repay the loan? *(2 marks)*

14. Financial Maths, 2ADV M1 2017 HSC 15b

- Anita opens a savings account. At the start of each month she deposits $\$X$ into the savings account. At the end of each month, after interest is added into the savings account, the bank withdraws \$2500 from the savings account as a loan repayment. Let M_n be the amount in the savings account after the n^{th} withdrawal.
- The savings account pays interest of 4.2% per annum compounded monthly.
- i. Show that after the second withdrawal the amount in the savings account is given by
- $$M_2 = X(1.0035^2 + 1.0035) - 2500(1.0035 + 1). \quad (2 \text{ marks})$$
- ii. Find the value of X so that the amount in the savings account is \$80 000 after the last withdrawal of the fourth year. *(3 marks)*

15. Financial Maths, 2ADV M1 2013 HSC 13d

A family borrows \$500 000 to buy a house. The loan is to be repaid in equal monthly instalments. The interest, which is charged at 6% per annum, is reducible and calculated monthly. The amount owing after n months, $\$A_n$, is given by

$$A_n = Pr^n - M(1 + r + r^2 + \dots + r^{n-1}) \quad (\text{DO NOT prove this})$$

where $\$P$ is the amount borrowed, $r = 1.005$ and $\$M$ is the monthly repayment.

- i. The loan is to be repaid over 30 years. Show that the monthly repayment is \$2998 to the nearest dollar. **(2 marks)**
- ii. Show that the balance owing after 20 years is \$270 000 to the nearest thousand dollars. **(1 mark)**
After 20 years the family borrows an extra amount, so that the family then owes a total of \$370 000. The monthly repayment remains \$2998, and the interest rate remains the same.
- iii. How long will it take to repay the \$370 000? **(2 marks)**

16. Financial Maths, 2ADV M1 2014 HSC 16b

At the start of a month, Jo opens a bank account and makes a deposit of \$500. At the start of each subsequent month, Jo makes a deposit which is 1% more than the previous deposit.

At the end of every month, the bank pays interest of 0.3% (per month) on the balance of the account.

- i. Explain why the balance of the account at the end of the second month is $\$500(1.003)^2 + \$500(1.01)(1.003)$. **(2 marks)**
- ii. Find the balance of the account at the end of the 60th month, correct to the nearest dollar. **(3 marks)**

17. Financial Maths, 2ADV M1 2019 HSC 16a

A person wins \$1 000 000 in a competition and decides to invest this money in an account that earns interest at 6% per annum compounded quarterly. The person decides to withdraw \$80 000 from this account at the end of every fourth quarter. Let A_n be the amount remaining in the account after the n th withdrawal.

- i. Show that the amount remaining in the account after the withdrawal at the end of the eighth quarter is $A_2 = 1\,000\,000 \times 1.015^8 - 80\,000(1 + 1.015^4)$. **(2 marks)**
- ii. For how many years can the full amount of \$80 000 be withdrawn? **(3 marks)**

18. Financial Maths, 2ADV M1 2018 HSC 16c

Kara deposits an amount of \$300 000 into an account which pays compound interest of 4% per annum, added to the account at the end of each year. Immediately after the interest is added, Kara makes a withdrawal for expenses for the coming year. The first withdrawal is $\$P$. Each subsequent withdrawal is 5% greater than the previous one.

Let $\$A_n$ be the amount in the account after the n th withdrawal.

- i. Show that $A_2 = 300\,000(1.04)^2 - P[(1.04) + (1.05)]$. **(1 mark)**
- ii. Show that $A_3 = 300\,000(1.04)^3 - P[(1.04)^2 + (1.04)(1.05) + (1.05)^2]$. **(1 mark)**
- iii. Show that there will be money in the account when

$$\left(\frac{105}{104}\right)^n < 1 + \frac{3000}{P} \quad (3 \text{ marks})$$

19. Financial Maths, 2ADV M1 2022 HSC 32

In a reducing-balance loan, an amount $\$P$ is borrowed for a period of n months at an interest rate of 0.25% per month, compounded monthly. At the end of each month, a repayment of $\$M$ is made. After the n th repayment has been made, the amount owing, $\$A_n$, is given by

$$A_n = P(1.0025)^n - M\left(1 + (1.0025)^1 + (1.0025)^2 + \dots + (1.0025)^{n-1}\right)$$

(Do NOT prove this.)

- a. Jane borrows \$200 000 in a reducing-balance loan as described.
The loan is to be repaid in 180 monthly repayments.
Show that $M = 1381.16$, when rounded to the nearest cent. **(2 marks)**
- b. After 100 repayments of \$1381.16 have been made, the interest rate changes to 0.35% per month.
At this stage, the amount owing to the nearest dollar is \$100 032. (Do NOT prove this.)
Jane continues to make the same monthly repayments.
For how many more months will Jane need to make full monthly payments of \$1381.16? **(3 marks)**
- c. The final payment will be less than \$1381.16.
How much will Jane need to pay in the final payment in order to pay off the loan? **(2 marks)**

20. Financial Maths, 2ADV M1 2008 HSC 9b

Peter retires with a lump sum of \$100 000. The money is invested in a fund which pays interest each month at a rate of 6% per annum, and Peter receives a fixed monthly payment $\$M$ from the fund. Thus the amount left in the fund after the first monthly payment is $\$(100\,500 - M)$.

- i. Find a formula for the amount, $\$A_n$, left in the fund after n monthly payments. **(2 marks)**
- ii. Peter chooses the value of M so that there will be nothing left in the fund at the end of the 12th year (after 144 payments). Find the value of M . **(3 marks)**

21. Financial Maths, 2ADV M1 2007 HSC 9c

Mr and Mrs Caine each decide to invest some money each year to help pay for their son's university education. The parents choose different investment strategies.

Mr Caine makes 18 yearly contributions of \$1000 into an investment fund. He makes his first contribution on the day his son is born, and his final contribution on his son's seventeenth birthday. His investment earns 6% compound interest per annum.

i. Find the total value of Mr Caine's investment on his son's eighteenth birthday. **(3 marks)**

Mrs Caine makes her contributions into another fund. She contributes \$1000 on the day of her son's birth, and increases her annual contribution by 6% each year. Her investment also earns 6% compound interest per annum.

ii. Find the total value of Mrs Caine's investment on her son's third birthday (just before she makes her fourth contribution). **(2 marks)**

iii. Mrs Caine also makes her final contribution on her son's seventeenth birthday. Find the total value of Mrs Caine's investment on her son's eighteenth birthday. **(1 mark)**

Worked Solutions

1. Financial Maths, 2ADV M1 2016 HSC 14b

i. $A_1 = 0.65 \times 100\,000 + 5000$

$$A_2 = 0.65 \times A_1 + 5000$$

$$= 0.65(0.65 \times 100\,000 + 5000) + 5000$$

... as required

ii. $A_2 = 0.65^2 \times 100\,000 + 0.65 \times 5000 + 5000$

$$A_3 = 0.65^3 \times 100\,000 + 0.65^2 \times 5000 + 0.65 \times 5000 + 5000$$

⋮

$$A_n = 0.65^n \times 100\,000 + 0.65^{n-1} \times 5000 + 0.65^{n-2} \times 5000 + \dots + 5000$$

$$= 0.65^n \times 100\,000 + 5000(1 + 0.65 + \dots + 0.65^{n-1})$$

$$\Rightarrow \text{GP where } a = 1, \quad r = 0.65$$

$$= 0.65^n \times 100\,000 + 5000 \left(\frac{1-r^n}{1-r} \right)$$

$$= 0.65^n \times 100\,000 + 5000 \frac{(1-0.65^n)}{0.35}$$

iii. $A_{14} = 0.65^{14} \times 100\,000 + 5000 \left(\frac{1-0.65^{14}}{0.35} \right)$

$$= 14\,491.70\dots$$

$$= 14\,500 \text{ (nearest 100)}$$

2. Financial Maths, 2ADV M1 2020 HSC 26

a. $A_1 = 60\,000(1.005) - 800 = \$59\,500$
 $A_2 = 59\,500(1.005) - 800 = \$58\,997.50$
 $A_3 = 58\,997.50(1.005) - 800 = \$58\,492.49$

b. Amount (not interest)
 $= 60\,000 - (3 \times 800)$
 $= \$57\,600$

$\therefore$ Interest earned in 3 months
 $= A_3 - 57\,600$
 $= 58\,492.49 - 57\,600$
 $= \$892.49$

c. $A_1 = 60\,000(1.005) - 800$
 $A_2 = [60\,000(1.005) - 800](1.005) - 800$
 $= 60\,000(1.005)^2 - 800(1.005 + 1)$
 $\vdots$
 $A_n = 60\,000(1.005)^n - 800(1 + 1.005 + \dots + 1.005^{n-1})$
 $A_{94} = 60\,000(1.005)^{94} - 800 \underbrace{(1 + 1.005 + \dots + 1.005^{93})}_{\text{GP where } a=1, r=1.005, n=94}$
 $= 60\,000(1.005)^{94} - 800 \left(\frac{1(1.005^{94} - 1)}{1.005 - 1} \right)$
 $= 60\,000(1.005)^{94} - 160\,000(1.005^{94} - 1)$
 $= \$187.85$

3. Financial Maths, 2ADV M1 2009 HSC 8b

i. Let L_n = the amount owing after n months
 $\text{Repayment} = M = \$2937$ and $r = \frac{9\%}{12} = 0.0075$ /month
 $\therefore L_1 = 350\,000(1 + r) - M$
 $= 350\,000(1.0075) - 2937$
 $= 349\,688$

$\therefore$ After 1 month, the amount owing is \$349 688

ii. After 12 repayments, Daniel owes \$346 095, and $r \downarrow 6\%$
 $\therefore r = \frac{6\%}{12} = 0.005$

Loan is repaid over the next 24 years. i.e. $A_n = 0$ when $n = 288$

$$A_n = 346\,095(1.005^n) - 1.005^{n-1}M - \dots - 1.005M - M$$
$$= 346\,095(1.005^n) - M(1 + 1.005 + \dots + 1.005^{n-1})$$

$$A_{288} = 346\,095(1.005^{288}) - M(1 + 1.005 + \dots + 1.005^{287}) = 0$$
$$\Rightarrow GP \text{ where } a = 1, \text{ and } r = 1.005$$

$$M \left(\frac{1(1.005^{288} - 1)}{1.005 - 1} \right) = 346\,095(1.005^{288})$$

$$M = \frac{1\,455\,529.832}{641.1158}$$
$$= 2270.31$$

$\therefore$ Monthly repayment is \$2270.31 (nearest cent)

iii. Given \$ M remains at \$2937, find n such that
 $A_n = 0$ (i.e. loan fully paid off)

$$\therefore 346\,095(1.005^n) - 2937 \left(\frac{1(1.005^n - 1)}{1.005 - 1} \right) = 0$$
$$346\,095(1.005^n) - 587\,400(1.005^n - 1) = 0$$
$$(346\,095 - 587\,400)(1.005^n) + 587\,400 = 0$$

$$241\,305(1.005^n) = 587\,400$$

$$\ln 1.005^n = \ln \left(\frac{587\,400}{241\,305} \right)$$

$$n = \frac{\ln 2.43426}{\ln 1.005}$$
$$= 178.37 \dots$$

MARKER'S COMMENT: Careless setting out and poor handwriting, especially where indexes were involved, was a major contributor to errors in this question.

♦♦ A poorly answered question.
MARKER'S COMMENT: Many students struggled to handle the exponential and logarithm calculations in this question.
ALGEBRA TIP: Dividing by $(1.005 - 1)$ in part (iii) is equivalent to multiplying by 200, and cleans up working calculations (see Worked Solutions).

$\therefore$ He will pay off the loan in 179 months (note the last payment will be a partial payment).

iv. Total paid at \$2937 per month

$$= 2937 \times 178.37 = \$523\,872.69$$

Total paid at \$2270.31 per month

$$= 2270.31 \times 288 = \$653,849.28$$

$\therefore$ The amount saved

$$= 653\,849.28 - 523\,872.69$$

$$= \$129\,976.59 \text{ (nearest cent)}$$

4. Financial Maths, 2ADV M1 2015 HSC 14c

i. $A_1 = 100\,000(1.006) - M$

$$A_2 = A_1(1.006) - M$$

$$= [100\,000(1.006) - M](1.006) - M$$

$$= 100\,000(1.006)^2 - M(1.006) - M$$

$$= 100\,000(1.006)^2 - M(1 + 1.006) \dots \text{ as required}$$

ii. $A_3 = 100\,000(1.006)^3 - M(1 + 1.006 + 1.006^2)$

$\vdots$

$$A_n = 100\,000(1.006)^n - M(1 + 1.006 + \dots + 1.006^{n-1})$$

$\Rightarrow$ Since $(1 + 1.006 + \dots + 1.006^{n-1})$ is a

GP with $a = 1$, $r = 1.006$

$$\therefore A_n = 100\,000(1.006)^n - M \left(\frac{a(r^n - 1)}{r - 1} \right)$$

$$= 100\,000(1.006)^n - M \left(\frac{1(1.006^n - 1)}{1.006 - 1} \right)$$

$$= 100\,000(1.006)^n - M \left(\frac{(1.006)^n - 1}{0.006} \right)$$

$\dots$ as required.

iii. If $M = 780$ and $n = 120$

$$A_{120} = 100\,000(1.006)^{120} - 780 \left(\frac{1.006^{120} - 1}{0.006} \right)$$

$$= 205\,001.80\dots - 780(175.0030\dots)$$

$$= 205\,001.80\dots - 136\,502.34\dots$$

$$= 68\,499.45\dots$$

$$= \$68\,500 \text{ (to nearest \$100)} \dots \text{ as required}$$

iv. After the one-off payment, amount owing = \$48 500

$$\therefore A_n = 48\,500(1.006)^n - 780 \left(\frac{1.006^n - 1}{0.006} \right)$$

where n is the number of months after the one-off payment.

Find n when $A_n = 0$

$$48\,500(1.006)^n - 780 \left(\frac{1.006^n - 1}{0.006} \right) = 0$$

$$48\,500(1.006)^n = 780 \left(\frac{1.006^n - 1}{0.006} \right)$$

$$48\,500(1.006)^n = 130\,000(1.006^n - 1)$$
$$= 130\,000(1.006)^n - 130\,000$$

$$81\,500(1.006)^n = 130\,000$$
$$1.006^n = \frac{130\,000}{81\,500}$$
$$n \times \ln 1.006 = \ln \frac{1300}{815}$$
$$n = \frac{\ln \frac{1300}{815}}{\ln 1.006}$$
$$= 78.055\dots$$

∴ The amount owing will be completely repaid after another 79 months.

5. Financial Maths, 2ADV M1 2021 HSC 29

a.

$$A_1 = 5000(1.03) + 1000$$
$$A_2 = [5000(1.03) + 1000](1.03) + 1000$$
$$= 5000(1.03)^2 + 1000(1.03) + 1000$$
$$A_3 = 5000(1.03)^3 + 1000(1.03)^2 + 1000(1.03) + 1000$$
$$= 8554.535$$
$$= \$8554.54$$

b. Megan’s 18th birthday onwards:

$$A_1 = 30\,025.83(1.03) - 2000$$
$$A_2 = [30\,025.83(1.03) - 2000](1.03) - 2000$$
$$= 30\,025.83(1.03)^2 - 2000(1.03) - 2000$$
$$\vdots$$
$$A_n = 30\,025.83(1.03)^n - 2000(1 + 1.03 + 1.03^2 + \dots + 1.03^{n-1})$$

Find n when $A_n = 0$:

$$30\,025.83(1.03)^n = 2000 \left[\frac{a(r^n - 1)}{r - 1} \right]$$
$$= 2000 \left(\frac{1.03^n - 1}{1.03 - 1} \right)$$
$$30\,025.83(1.03)^n = 66\,666.67(1.03)^n - 66\,666.67$$
$$36\,640.84(1.03)^n = 66\,666.67$$
$$1.03^n = \frac{66\,666.67}{36\,640.84}$$
$$n \ln 1.03 = \ln \left(\frac{66\,666.67}{36\,640.84} \right)$$
$$n = \frac{\ln \left(\frac{66\,666.67}{36\,640.84} \right)}{\ln 1.03}$$
$$= 20.24\dots$$

∴ Megan can make 20 withdrawals of \$2000.

6. Financial Maths, 2ADV M1 2023 HSC 25

- a. $A_1 = 10\,000(1.004) - M$
 $A_2 = A_1(1.004) - M$
 $= [(10\,000(1.004) - M)](1.004) - M$
 $= 10\,000(1.004)^2 - M(1.004) - M$
- b. $A_3 = 10\,000(1.004)^3 - M(1.004)^2 - M(1.004) - M$
 $\vdots$
 $A_n = 10\,000(1.004)^n - M(1.004)^{n-1} - \dots - M(1.004) - M$
 $= 10\,000(1.004)^n - \underbrace{M(1 + 1.004 + 1.004^2 + \dots + 1.004^{n-1})}_{\text{GP where } a=1, r=1.004, n=n}$
 $A_n = 10\,000(1.004)^n - M\left(\frac{1.004^n - 1}{1.004 - 1}\right)$
 $= 10\,000(1.004)^n - M\left(\frac{1.004^n - 1}{0.004}\right)$
 $= 10\,000(1.004)^n - 250M(1.004^n - 1)$
 $= 10\,000(1.004)^n - 250M(1.004)^n + 250M$
 $= (10\,000 - 250M)(1.004)^n + 250M$
- c. Find M when $A_{100} = 0$:
 $A_{100} = (10\,000 - 250M)(1.004)^{100} + 250M$
 $0 = 10\,000(1.004)^{100} - 250M(1.004)^{100} + 250M$
 $250M(1.004^{100} - 1) = 10\,000(1.004)^{100}$
 $M = \frac{10\,000(1.004)^{100}}{250(1.004^{100} - 1)}$
 $= 121.527\dots$

 $\therefore M_{\max} = \$121.52$

♦ Mean mark (b) 44%.

Mean mark (c) 53%.

7. Financial Maths, 2ADV M1 2024 HSC 24

- a. $r = \frac{6\%}{12} = 0.5\%$ per month
Let $A_n =$ Amount after n months
 $A_1 = 80(1.005)$
 $A_2 = [80(1.005) + 80](1.005)$
 $= 80(1.005)^2 + 80(1.005)$
 $\vdots$
 $A_n = 80(1.005)^n + 80(1.005)^{n-1} + \dots + 80(1.005)$
 $= 80(\underbrace{1.005 + 1.005^2 + \dots + 1.005^n}_{\text{GP where } a=1.005, r=1.005})$
 $= 80\left(\frac{1.005(1.005^n - 1)}{1.005 - 1}\right)$

After 24 months:
 $A_{24} = 80\left(\frac{1.005(1.005^{24} - 1)}{1.005 - 1}\right)$
 $= \$2044.73$ (nearest cent)
- b. $FV = 80 \times A$
 $2044.73 = 80 \times A$
 $A = \frac{2044.73}{80}$
 $= 25.5591\dots$
 $= 25.559$ (3 d.p.)

♦♦ Mean mark (a) 38%.

Mean mark (b) 51%.

8. Financial Maths, 2ADV M1 2010 HSC 9a

i. $P_1 = 500(1.005)$

$$P_2 = 500(1.005^2) + 500(1.005^1)$$

$$P_3 = 500(1.005^3 + 1.005^2 + 1.005)$$

$$\vdots$$

$$P_{240} = 500(1.005 + 1.005^2 + 1.005^3 \dots + 1.005^{240})$$

$\Rightarrow$ GP where $a = 1.005$, and $r = 1.005$

$$\begin{aligned} P_{240} &= 500 \left(\frac{a(r^n - 1)}{r - 1} \right) \\ &= 500 \left(\frac{1.005(1.005^{240} - 1)}{1.005 - 1} \right) \\ &= 100\,000 [1.005(1.005^{240} - 1)] \\ &= 232\,175.55 \end{aligned}$$

$\therefore$ The value of Chris' investment after 240 months is \$232 175.55 ... as required

MARKER'S COMMENT: Common errors included using $r = 1.05$, and taking the first term of the GP as 1 instead of 1.005 (note that the \$500 goes in at the start of the month and earns interest before it is included in $\$P_n$).

ii. (1) After 1 month, $A_1 = P(1.005) - 2000$

$$\begin{aligned} A_2 &= A_1(1.005) - 2000 \\ &= [P(1.005) - 2000](1.005) - 2000 \\ &= P(1.005^2) - 2000(1.005) - 2000 \\ &= P(1.005^2) - 2000(1 + 1.005) \end{aligned}$$

$$\vdots$$

$$\begin{aligned} A_n &= P(1.005^n) - 2000(1 + 1.005 + \dots + 1.005^{n-1}) \\ &\Rightarrow \text{GP where } a = 1 \text{ and } r = 1.005 \end{aligned}$$

$$\begin{aligned} A_n &= P(1.005^n) - 2000 \left(\frac{1(1.005^n - 1)}{1.005 - 1} \right) \\ &= P(1.005^n) - 400\,000(1.005^n - 1) \\ &= P(1.005^n) - 400\,000(1.005^n) + 400\,000 \\ &= (P - 400\,000) \times 1.005^n + 400\,000 \dots \text{as required} \end{aligned}$$

IMPORTANT: At the end of the month, $\$P$ earns interest for the month *BEFORE* any withdrawal is made. Many students mistakenly had $\$A_1 = (P - 2000)(1.005)$.

ii. (2) Find n such that $A_n \leq 0$

Since $P = 232\,175.55$,

$$(232\,175.55 - 400\,000)(1.005^n) + 400\,000 \leq 0$$

$$167\,824.45(1.005^n) \geq 400\,000$$

$$1.005^n$$

♦ Mean mark 38%

$$\begin{aligned}
&\geq \frac{400\,000}{167\,824.45} \\
n \ln 1.005 &\geq \ln 2.383443 \\
n &\geq \frac{\ln 2.383443}{\ln 1.005} \\
n &\geq 174.14 \quad (\text{to 2 d.p.})
\end{aligned}$$

$\therefore$ There will be money left in the account for 175 months.

9. Financial Maths, 2ADV M1 2011 HSC 8c

$$\text{i.} \quad P_1 = P \times r = P \times \left(1 + \frac{6\%}{12}\right) = 100(1.005)$$

$$\begin{aligned}
P_2 &= P_1(1.005) + 100(1.005^1) \\
&= 100(1.005^2) + 100(1.005^1) \\
&= 100(1.005^2 + 1.005)
\end{aligned}$$

$$\begin{aligned}
P_3 &= (1.005)[100(1.005^2 + 1.005)] + 100(1.005) \\
&= 100(1.005 + 1005^2 + 1.005^3)
\end{aligned}$$

$\vdots$

$$\begin{aligned}
P_{420} &= 100(1.005 + 1.005^2 + 1.005^3 \dots + 1.005^{420}) \\
&\Rightarrow \text{GP where } a = 1.005, \quad r = 1.005
\end{aligned}$$

$$\begin{aligned}
P_{420} &= 100 \left(\frac{a(r^n - 1)}{r - 1} \right) \\
&= 100 \left(\frac{1.005(1.005^{420} - 1)}{1.005 - 1} \right) \\
&= 20\,000(1.005(1.005^{420} - 1)) \\
&= \$143\,183.39
\end{aligned}$$

$\therefore$ The final value of Jules's superannuation is
\$143 183 (to the nearest dollar) ... as required

ii. (1) After 1 month, $A_1 = 29\,227(1.005) + M(1.005)$

$$\begin{aligned}
A_2 &= A_1(1.005) + M(1.005) \\
&= [29\,227(1.005) + M(1.005)](1.005) + M(1.005) \\
&= 29\,227(1.005^2) + M(1.005^2) + M(1.005) \\
&= 29\,227(1.005^2) + M(1.005 + 1.005^2) \dots \text{as required}
\end{aligned}$$

ii. (2) Find $\$M$ such that $A_n = \$800\,000$ after 240 months.

$$\begin{aligned}
A_{240} &= 29\,227(1.005^{240}) + M(1.005 + 1.005^2 + \dots + 1.005^{240}) \\
&\Rightarrow GP \text{ where } a = 1.005, \text{ and } r = 1.005
\end{aligned}$$

$$800\,000 = 29\,227(1.005^{240}) + M \left(\frac{1.005(1.005^{240} - 1)}{1.005 - 1} \right)$$

$$M \left(\frac{1.005(1.005^{240} - 1)}{1.005 - 1} \right) = 800\,000 - 29\,227(1.005^{240})$$

M

MARKER'S COMMENT: Common errors in this part included having the first term of the GP as 1 instead of 1.005 (note that the \$100 goes in at the start of the month and earns interest before it is included in $\$P_n$).

◆ Mean mark 34% for part (ii)(1)

◆ Mean mark 49%

$$= \frac{703\,252.65}{464.3511}$$

$$= 1514.484.$$

$$\therefore M = \$1514.48 \quad (\text{to the nearest cent})$$

10. Financial Maths, 2ADV M1 2012 HSC 15c

$$\text{i. } A_1 = 360\,000 \left(1 + \frac{6\%}{12} \right) - M$$

$$= 360\,000(1.005) - M$$

$$A_2 = [360\,000(1.005) - M](1.005) - M$$

$$= 360\,000(1.005^2) - M(1.005) - M$$

$$= 360\,000(1.005^2) - M(1 + 1.005)$$

$$\text{ii. } A_n = 360\,000(1.005^n) - M(1 + 1.005^1 + \dots + 1.005^{n-1})$$

$$\text{When } n = 300, A_n = 0$$

$$0 = 360\,000(1.005^{300}) - M(1 + 1.005^1 + \dots + 1.005^{299})$$

$$360\,000(1.005^{300}) = M \left(\frac{a(r^n - 1)}{r - 1} \right)$$

$$M \left(\frac{1(1.005^{300} - 1)}{1.005 - 1} \right) = 360\,000(1.005^{300})$$

$$\therefore M = \left(\frac{1\,607\,389.13}{692.994} \right)$$

$$\approx 2319.50 \quad \dots \text{ as required}$$

$$\text{iii. Find } n \text{ such that } \$A_n < \$180\,000$$

$$360\,000(1.005^n) - 2319.50 \left(\frac{1.005^n - 1}{1.005 - 1} \right) < 180\,000$$

$$360\,000(1.005^n) - 463\,900(1.005^n - 1) < 180\,000$$

$$-103\,900(1.005^n) + 463\,900 < 180\,000$$

$$103\,900(1.005^n) > 283\,900$$

$$1.005^n > \frac{283\,900}{103\,900}$$

$$n(\ln 1.005) > \ln \left(\frac{283\,900}{103\,900} \right)$$

$$n > \frac{1.005193}{0.0049875}$$

$$n > 201.54$$

$$\therefore \text{After 202 months, } \$A_n < \$180\,000.$$

11. Financial Maths, 2ADV M1 2004 HSC 7c

$$\text{i. } A = P \left(1 + \frac{r}{100} \right)^n$$

$$P = \$80$$

$$n = 300 \quad \text{months}$$

$$r = 6\% \quad \text{per annum}$$

$$= 0.5\% \quad \text{per month}$$

$$\therefore A = 80 \left(1 + \frac{0.5}{100} \right)^{300}$$

$$= 80(1.005)^{300}$$

$$= 357.1975\dots$$

$$= \$357.20 \quad (\text{nearest cent})$$

$$\text{ii. } A_1 = 80(1.005)^{300}$$

$$A_2 = 80(1.005)^{299}$$

$$A_3 = 80(1.005)^{298}$$

$$\vdots$$

$$A_{300} = 80(1.005)^1$$

$$\therefore \text{Final value of fund}$$

$$= 80(1.005)^{300} + 80(1.005)^{299} + \dots + 80(1.005)^1$$

$$= 80[1.005^{300} + 1.005^{299} + \dots + 1.005^1]$$

$$= 80[1.005^1 + 1.005^2 + \dots + 1.005^{300}]$$

$$\Rightarrow \text{GP where } a = 1.005, r = 1.005, n = 300$$

$$= 80 \left[\frac{a(r^n - 1)}{r - 1} \right]$$

$$= 80 \left[\frac{1.005(1.005^{300} - 1)}{1.005 - 1} \right]$$

$$= 55\,716.714\dots$$

$$= \$55\,716.71 \quad \text{to the nearest cent.}$$

◆ Mean mark 38%
MARKER'S COMMENT: Challenging calculations using logarithms are common in this topic. A high percentage of students consistently struggle in this area.

12. Financial Maths, 2ADV M1 2005 HSC 8c

i. $A_1 = (3 \times 10^6)(1.12) - (4.8 \times 10^5)$

$$\begin{aligned} A_2 &= A_1(1.12) - (4.8 \times 10^5) \\ &= [(3 \times 10^6)(1.12) - (4.8 \times 10^5)](1.12) - (4.8 \times 10^5) \\ &= (3 \times 10^6)(1.12)^2 - (4.8 \times 10^5)(1.12) - (4.8 \times 10^5) \\ &= (3 \times 10^6)(1.12)^2 - (4.8 \times 10^5)(1 + 1.12) \dots \text{as required} \end{aligned}$$

ii. Show $A_n = 10^6[4 - (1.12)^n]$

$$\begin{aligned} A_3 &= (3 \times 10^6)(1.12)^3 - (4.8 \times 10^5)(1 + 1.12 + 1.12^2) \\ &\vdots \\ A_n &= (3 \times 10^6)(1.12)^n - (4.8 \times 10^5)(1 + 1.12 + \dots + 1.12^{n-1}) \\ &\Rightarrow \text{Noting that } (1 + 1.12 + \dots + 1.12^{n-1}) \text{ is a} \\ &\quad \text{GP where } a = 1, r = 1.12 \end{aligned}$$

$$\begin{aligned} A_n &= (3 \times 10^6)(1.12)^n - (4.8 \times 10^5) \left(\frac{1(1.12^n - 1)}{1.12 - 1} \right) \\ &= (3 \times 10^6)(1.12)^n - (4.8 \times 10^5) \left(\frac{1(1.12^n - 1)}{0.12} \right) \\ &= (3 \times 10^6)(1.12)^n - (4 \times 10^6)(1.12^n - 1) \\ &= 10^6[3(1.12)^n - 4(1.12^n - 1)] \\ &= 10^6[3(1.12)^n - 4(1.12^n) + 4] \\ &= 10^6[4 - (1.12)^n] \dots \text{as required} \end{aligned}$$

iii. Find n such that $A_n = 0$

$$\begin{aligned} 10^6[4 - (1.12)^n] &= 0 \\ 4 - (1.12)^n &= 0 \\ 1.12^n &= 4 \\ n \times \ln 1.12 &= \ln 4 \\ n &= \frac{\ln 4}{\ln 1.12} \\ &= 12.23 \dots \text{ years.} \end{aligned}$$

$\therefore$ The final repayment will be made
in 2017, the thirteenth year.

13. Financial Maths, 2ADV M1 2006 HSC 8b

i. $A_n = 200\,000r^n - M(1 + r + r^2 + \dots + r^{n-1})$

Find M such that $A_n = 0$ when $n = 300$

$$0 = 200\,000(1.006^{300}) - M(1 + 1.006 + \dots + 1.006^{299})$$

$\Rightarrow$ Note that $(1 + 1.006 + \dots + 1.006^{299})$ is a

GP where $a = 1, r = 1.006, n = 300$

$$0 = 200\,000(1.006^{300}) - M \left(\frac{1(1.006^{300} - 1)}{1.006 - 1} \right)$$

$$M \left(\frac{(1.006^{300} - 1)}{0.006} \right) = 200\,000(1.006^{300})$$

$$\begin{aligned} M &= \frac{1\,203\,439.36 \dots}{836.199 \dots} \\ &= 1439.177 \dots \\ &= \$1439 \text{ (nearest dollar)} \end{aligned}$$

$\therefore$ The minimum monthly repayment is \$1439.

ii. If $M = 2800$, find n when $A_n = 0$

$$0 = 200\,000(1.006^n) - 2800 \left(\frac{(1.006^n - 1)}{0.006} \right)$$

$$0 = 1200(1.006^n) - 2800(1.006^n) + 2800$$

$$0 = -1600(1.006^n) + 2800$$

$$1600(1.006^n) = 2800$$

$$1.006^n = \frac{2800}{1600}$$

$$n \times \ln 1.006 = \ln 1.75$$

$$n = \frac{\ln 1.75}{\ln 1.006}$$

$$= 93.54 \dots$$

$$= 94 \text{ months (nearest month)}$$

$\therefore$ It will take Joe 94 months to repay the loan.

14. Financial Maths, 2ADV M1 2017 HSC 15b

i. $M_1 = X\left(1 + \frac{4.2}{12 \times 100}\right) - 2500$

$$= X(1.0035) - 2500$$
$$M_2 = X(1.0035) + M_1(1.0035) - 2500$$
$$= X(1.0035) + (1.0035)[X(1.0035) - 2500] - 2500$$
$$= X(1.0035) + X(1.0035^2) - 2500(1.0035) - 2500$$
$$= X(1.0035^2 + 1.0035) - 2500(1.0035 + 1)$$

♦ Mean mark part (ii) 48%.

ii. $M_3 = X(1.0035^3 + \dots + 1.0035) - 2500(1.0035^2 + 1.0035 + 1)$

$\vdots$

$$M_n = X(1.0035^n + \dots + 1.0035) - 2500(1.0035^{n-1} + \dots + 1)$$

Find X such that $M_n = 80\,000$ when $n = 48$

$$80\,000 = X(1.0035^{48} + \dots + 1.0035) - 2500(1.0035^{47} + \dots + 1)$$

Using $S_n = \frac{a(r^n - 1)}{r - 1}$

$$80\,000 = X\left[\frac{1.0035(1.0035^{48} - 1)}{1.0035 - 1}\right] - 2500\left[\frac{1(1.0035^{48} - 1)}{1.0035 - 1}\right]$$
$$80\,000 = X(52.351\dots) - 130\,421.20\dots$$
$$\therefore X = \frac{80\,000 + 130\,421.20\dots}{52.351\dots}$$
$$= \$4019.42 \text{ (nearest cent)}$$

15. Financial Maths, 2ADV M1 2013 HSC 13d

i. Find $\$M$ such that the loan is repaid over 30 years.

$$n = 30 \times 12 = 360 \text{ periods} \quad r = 1 + \frac{6}{12}\% = 1.005$$
$$A_{360} = 500\,000(1.005^{360}) - M(1 + 1.005 + \dots + 1.005^{359}) = 0$$
$$\Rightarrow GP \text{ where } a = 1, \quad r = 1.005, \quad n = 360$$
$$M\left(\frac{1(1.005^{360} - 1)}{1.005 - 1}\right) = 500\,000(1.005^{360})$$
$$M(1004.515) = 3\,011\,287.61$$
$$M = 2997.75$$

$$\therefore \$M = \$2998 \text{ (nearest dollar) } \dots \text{ as required}$$

ii. Find $\$A_n$ after 20 years $\Rightarrow n = 20 \times 12 = 240$

$$A_{240} = 500\,000(1.005^{240}) - 2998(1 + 1.005 + \dots + 1.005^{239})$$
$$= 1\,655\,102.24 - 2998\left(\frac{1(1.005^{240} - 1)}{1.005 - 1}\right)$$
$$= 269\,903.63$$
$$= 270\,000 \text{ (nearest thousand) } \dots \text{ as required}$$

MARKER'S COMMENT: Within the GP formula, many students incorrectly wrote the last term as 1.005^{240} rather than 1.005^{239} . Note $T_n = ar^{n-1}$.

iii. Loan = \$370 000

Find n such that $\$A_n = 0, \quad M = \2998

$$A_n = 370\,000(1.005^n) - 2998(1 + 1.005 + \dots + 1.005^{n-1}) = 0$$
$$370\,000(1.005^n) = 2998\left(\frac{1(1.005^n - 1)}{1.005 - 1}\right)$$
$$370\,000(1.005^n) = 599\,600(1.005^n - 1)$$
$$229\,600(1.005^n) = 599\,600$$
$$\ln 1.005^n = \ln\left(\frac{599\,600}{229\,600}\right)$$
$$n = \frac{\ln 2.6115}{\ln 1.005}$$
$$n = 192.4$$

$$\therefore \text{The loan will be repaid after 193 months.}$$

♦♦ Mean mark 33%
COMMENT: Another good examination of working with logarithms. Students should understand why they must 'round up' their answer in this question.

16. Financial Maths, 2ADV M1 2014 HSC 16b

i. Let A_n = balance at end of month n

$$\begin{aligned} A_1 &= 500(1.003) \\ A_2 &= A_1(1.003) + 500(1.01)(1.003) \\ &= 500(1.003)(1.003) + 500(1.01)(1.003) \\ &= 500(1.003)^2 + 500(1.01)(1.003) \end{aligned}$$

ii. Need to find A_{60}

$$\begin{aligned} A_3 &= A_2(1.003) + 500(1.01)^2(1.003) \\ &= \left[500(1.003)^2 + 500(1.01)(1.003) \right] (1.003) \\ &\quad + 500(1.01)^2(1.003) \\ &= 500(1.003)^3 + 500(1.003)^2(1.01) + 500(1.003)(1.01)^2 \\ &= 500 \left[1.003(1.01)^2 + (1.003)^2(1.01) + (1.003)^3 \right] \end{aligned}$$

$$\begin{aligned} &\vdots \\ A_{60} &= 500 \left[1.003(1.01)^{59} + (1.003)^2(1.01)^{58} + \dots \right. \\ &\quad \left. \dots + (1.003)^{60} \right] \end{aligned}$$

$$\Rightarrow \text{GP where } a = 1.003(1.01)^{59}, \quad r = \frac{T_2}{T_1} = \frac{1.003}{1.01}$$

$$\begin{aligned} \therefore A_{60} &= 500 \left[\frac{a(r^n - 1)}{r - 1} \right] \\ &= 500 \left[\frac{1.003(1.01)^{59} \left(\left(\frac{1.003}{1.01} \right)^{60} - 1 \right)}{\frac{1.003}{1.01} - 1} \right] \\ &= 500 \left[\frac{1.8041057(- 0.3411697)}{- 0.00693069} \right] \\ &= 44\,404.396\dots \\ &= \$44\,404 \text{ (nearest dollar)} \end{aligned}$$

$\therefore$ At the end of 60 months, the balance would be \$44 404.

◆ Mean mark 38%

17. Financial Maths, 2ADV M1 2019 HSC 16a

i. 6% p.a. = 1.5% per quarter

$$\begin{aligned} A_1 &= 1\,000\,000 \times 1.015^4 - 80\,000 \\ A_2 &= A_1 \times 1.015^4 - 80\,000 \\ &= (1\,000\,000 \times 1.015^4 - 80\,000) \times 1.015^4 - 80\,000 \\ &= 1\,000\,000 \times 1.015^8 - 80\,000 \times 1.015^4 - 80\,000 \\ &= 1\,000\,000 \times 1.015^8 - 80\,000(1 + 1.015^4) \end{aligned}$$

ii. $A_3 = 1\,000\,000 \times 1.015^{12} - 80\,000(1 + 1.015^4 + 1.015^8)$

$$\begin{aligned} &\vdots \\ A_n &= 1\,000\,000 \times 1.015^{4n} - 80\,000 \underbrace{(1 + 1.015^4 + \dots + 1.015^{4n-4})}_{\text{GP where } a=1, \, r=1.015^4} \\ &= 1\,000\,000 \times 1.015^{4n} - 80\,000 \left[\frac{1(1.015^{4n} - 1)}{1.015^4 - 1} \right] \\ &= 1\,000\,000 \times 1.015^{4n} - 1\,303\,706(1.015^{4n} - 1) \end{aligned}$$

Find n when $A_n = 0$:

◆◆◆ Mean mark part (ii) 19%.

$$\begin{aligned} 0 &= 1\,000\,000 \times 1.015^{4n} - 1\,303\,706 \times 1.015^{4n} + 1\,303\,706 \\ -1\,303\,706 &> -303\,706 \times 1.015^{4n} \\ 1.015^{4n} &> \frac{1\,303\,706}{303\,706} \\ 4n &> \frac{\ln\left(\frac{1\,303\,706}{303\,706}\right)}{\ln 1.015} \\ &> 97.853\dots \\ n &> 24.46\dots \end{aligned}$$

$\therefore$ Full amount can be drawn for 24 years.

18. Financial Maths, 2ADV M1 2018 HSC 16c

i. $A_1 = 300\,000(1.04) - P$

$$\begin{aligned} A_2 &= [300\,000(1.04) - P](1.04) - P(1.05) \\ &= 300\,000(1.04)^2 - P(1.04) - P(1.05) \\ &= 300\,000(1.04)^2 - P[1.04 + 1.05] \end{aligned}$$

♦ Mean mark 47%.

ii. $A_3 = [300\,000(1.04)^2 - P(1.04 + 1.05)](1.04) - P(1.05)^2$

$$\begin{aligned} &= 300\,000(1.04)^3 - P(1.04)^2 - P(1.04)(1.05) - P(1.05)^3 \\ &= 300\,000(1.04)^3 - P[(1.04)^2 + P(1.04)(1.05) + (1.05)^2] \end{aligned}$$

iii. $A_4 = 300\,000(1.04)^4 - P[(1.04)^3 + (1.04)^2(1.05) + \dots + (1.05)^3]$

⋮

$$A_n = 300\,000(1.04)^n - P \underbrace{\left[(1.04)^{n-1} + (1.04)^{n-2}(1.05) + \dots + (1.04)(1.05)^{n-2} + (1.05)^{n-1} \right]}_{\text{GP, } a=(1.04)^{n-1}, r=\frac{1.05}{1.04}}$$

Money in account when $A_n > 0$:

♦♦♦ Mean mark part (iii) 8%.

$$\begin{aligned} \frac{300\,000(1.04)^n}{P} &> (1.04)^{n-1} \left[\frac{\left(\frac{1.05}{1.04}\right)^n - 1}{\left(\frac{1.05}{1.04}\right) - 1} \right] \\ \frac{300\,000}{P} &> \frac{1}{1.04} \left[\frac{\left(\frac{1.05}{1.04}\right)^n - 1}{\left(\frac{1.05}{1.04}\right) - 1} \right] \\ \frac{300\,000}{P} &> \frac{\left(\frac{1.05}{1.04}\right)^n - 1}{1.05 - 1.04} \\ \frac{3000}{P} &> \left(\frac{1.05}{1.04}\right)^n - 1 \\ \frac{3000}{P} + 1 &> \left(\frac{105}{104}\right)^n \quad \left(\text{since } \left(\frac{1.05}{1.04}\right)^n = \left(\frac{105}{104}\right)^n\right) \end{aligned}$$

19. Financial Maths, 2ADV M1 2022 HSC 32

a. Show $M = \$1381.16$

$$\begin{aligned} A_n &= P(1.0025)^n - M \left(1 + (1.0025)^1 + (1.0025)^2 + \dots + (1.0025)^{n-1} \right) \\ 0 &= 200\,000(1.0025)^{180} - \underbrace{M \left(1 + (1.0025)^1 + (1.0025)^2 + \dots + (1.0025)^{179} \right)}_{\text{GP where } a=1, r=1.0025, n=180} \\ 0 &= 200\,000(1.0025)^{180} - M \left(\frac{1(1.0025^{180} - 1)}{1.0025 - 1} \right) \\ M \left(\frac{1.0025^{180} - 1}{1.0025 - 1} \right) &= 200\,000(1.0025)^{180} \\ \therefore M &= 200\,000(1.0025)^{180} \div \frac{1.0025^{180} - 1}{1.0025 - 1} \\ &= 1381.163\dots \\ &= \$1381.16 \quad \dots \text{ as required} \end{aligned}$$

b. $P = \$100\,032$, $r = 1.0035$ and $M = \$1381.16$

Find n when $A_n = 0$:

♦♦ Mean mark part (b) 38%.

$$\begin{aligned} A_n &= P(1.0035)^n - 1381.16 \left(1 + (1.0035)^1 + (1.0035)^2 + \dots + (1.0035)^{n-1} \right) \\ 0 &= 100\,032(1.0035)^n - 1381.16 \left(\frac{1.0035^n - 1}{1.0035 - 1} \right) \\ 0 &= 100\,032(1.0035)^n - \frac{1381.16}{0.0035}(1.0035^n - 1) \\ &= 100\,032(1.0035)^n - 394\,617(1.0035)^n - 394\,617 \\ 294\,585(1.0035)^n &= 394\,617 \\ 1.0035^n &= \frac{394\,617}{294\,585} \\ n &= \frac{\ln\left(\frac{394\,617}{294\,585}\right)}{\ln 1.0035} \\ &= 83.674\dots \\ &= 83 \text{ more months with full payment} \end{aligned}$$

c. Find A_{83} :

♦♦♦ Mean mark part (c) 14%.

$$\begin{aligned} A_{83} &= 100\,032(1.0035)^{83} - 1381.16 \left(\frac{1.0035^{83} - 1}{1.0035 - 1} \right) \\ &= 928.291\dots \end{aligned}$$

Interest will be added for the last month:

$$\begin{aligned} \therefore \text{Final payment} &= 928.291\dots \times 1.0035 \\ &= \$931.54 \end{aligned}$$

20. Financial Maths, 2ADV M1 2008 HSC 9b

i. $r = \left(1 + \frac{0.06}{12}\right) = 1.005$

$$A_1 = (100\,500 - M)$$

$$= 100\,000(1.005)^1 - M$$

$$A_2 = A_1(1.005) - M$$

$$= [100\,000(1.005^1) - M](1.005) - M$$

$$= 100\,000(1.005^2) - M(1.005) - M$$

$$= 100\,000(1.005^2) - M(1 + 1.005)$$

$$A_3 = 100\,000(1.005^3) - M(1 + 1.005^1 + 1.005^2)$$

$$\vdots$$

$$A_n = 100\,000(1.005^n) - M(1 + 1.005 + \dots + 1.005^{n-1})$$

$$= 100\,000(1.005^n) - M\left(\frac{a(r^n - 1)}{r - 1}\right)$$

$$= 100\,000(1.005^n) - M\left(\frac{1(1.005^n - 1)}{1.005 - 1}\right)$$

$$= 100\,000(1.005^n) - M\left(\frac{1.005^n - 1}{0.005}\right)$$

ii. Find M such that $\$A_n = 0$ when $n = 144$

$$A_{144} = 100\,000(1.005)^{144} - M\left(\frac{1.005^{144} - 1}{0.005}\right) = 0$$

$$M\left(\frac{1.005^{144} - 1}{0.005}\right) = 100\,000(1.005^{144})$$

$$M(1.005^{144} - 1) = 500(1.005^{144})$$

$$M = \frac{500(1.005^{144})}{1.005^{144} - 1}$$

$$= 975.85$$

$\therefore M = \$975.85$ (nearest cent).

MARKER'S COMMENT: Students who developed this answer from writing the calculations of A_1, A_2, A_3 to then generalising for A_n were the most successful.

21. Financial Maths, 2ADV M1 2007 HSC 9c

i. Let A_n = value of the investment after n years

$$\begin{aligned}A_1 &= 1000(1.06) \\A_2 &= A_1(1.06) + 1000(1.06) \\&= [1000(1.06)](1.06) + 1000(1.06) \\&= 1000(1.06)^2 + 1000(1.06) \\A_3 &= A_2(1.06) + 1000(1.06) \\&= 1000(1.06)^3 + 1000(1.06)^2 + 1000(1.06) \\&= 1000[1.06 + 1.06^2 + 1.06^3] \\&\vdots \\A_n &= 1000[1.06 + 1.06^2 + \dots + 1.06^n] \\&\text{Note that } (1.06 + 1.06^2 + \dots + 1.06^n) \text{ is a} \\&\text{GP where } a = 1.06 \text{ and } r = 1.06\end{aligned}$$

$$\therefore A_n = 1000 \left[\frac{a(r^n - 1)}{r - 1} \right]$$

The son’s 18th birthday occurs when $n = 18$

$$\begin{aligned}\therefore A_{18} &= 1000 \left[\frac{1.06(1.06^{18} - 1)}{1.06 - 1} \right] \\&= 1000 \times 32.7599\dots \\&= 32\,759.991\dots \\&= \$32\,760 \quad (\text{nearest \$})\end{aligned}$$

$\therefore$ The value of Mr Caine’s investment is \$32 760.

ii. Let V_n = value of investment after n years

$$\begin{aligned}V_1 &= 1000(1.06) \\V_2 &= V_1(1.06) + 1000(1.06)(1.06) \\&= 1000(1.06)^2 + 1000(1.06)^2 \\&= 2000(1.06)^2 \\V_3 &= V_2(1.06) + 1000(1.06)^3 \\&= 2000(1.06)^3 + 1000(1.06)^3 \\&= 3000(1.06)^3 \\&= 3573.048\dots \\&= \$3573 \quad (\text{nearest dollar})\end{aligned}$$

iii. Continuing the pattern

$$\begin{aligned}V_4 &= 4000(1.06)^4 \\&\vdots \\V_n &= n \times 1000(1.06)^n \\ \therefore V_{18} &= 18 \times 1000 \times (1.06)^{18} \\&= 51\,378.104\dots \\&= \$51\,378 \quad (\text{nearest dollar})\end{aligned}$$

$\therefore$ The value of Mrs Caine’s investment on her son’s 18th birthday will be \$51 378.